Hybrid Recommendation System for Anime and Movies Using Machine Learning

Kritika kumari
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai
Chennai
25MCA1048

Harsh Pratap Singh
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai
Chennai
25MCA1027

***Abstarct*— In the era of digital streaming, users are often overwhelmed by the sheer volume of entertainment content available, a phenomenon known as "choice overload." This project presents a Hybrid Recommendation System that integrates Collaborative Filtering (CF) and Content-Based Filtering (CBF) to recommend both Anime and Movies. While Content-Based methods suggest items similar to what a user has liked in the past, Collaborative Filtering relies on the preferences of similar users. By combining these two approaches using Machine Learning, this system aims to overcome common limitations such as the "cold start" problem (new users/items) and data sparsity. The system utilizes metadata such as genres, synopsis, and cast, along with user rating history, to generate personalized, accurate, and diverse recommendations. The input for the analyzer is a structured dataset of user ratings and media metadata. The main goals of this paper are to design the framework, analyze the effectiveness of hybrid techniques, and visualize recommendation clusters.**

Keywords*— *Recommendation System, Machine Learning, Hybrid Filtering, Collaborative Filtering, Content-Based Filtering, Anime, Movies.

I. Introduction

The entertainment industry has witnessed a paradigm shift from traditional television to on-demand streaming platforms. With databases containing thousands of anime titles and movies, finding relevant content is a significant challenge for users. These platforms generate massive amounts of data regarding user preferences, watch history, and ratings. However, navigating this unstructured and vast data manually is impossible. Users often spend more time searching for content than watching it. Therefore, there is a growing need for automated systems capable of filtering information and predicting items a user might like. Machine Learning (ML) techniques offer robust solutions for building recommendation engines. Traditional approaches, such as standalone Collaborative Filtering or Content-Based Filtering, have inherent flaws. Collaborative Filtering fails when a new user joins (cold start), while Content-Based Filtering fails to recommend novel items outside a user's niche. This project proposes a **Hybrid approach**, which is particularly suitable for a mixed catalog of Anime and Movies, ensuring that users receive recommendations that are both semantically similar to their interests and validated by the community..

A. Research Goal

The primary goal of this research is to design and evaluate a **Machine Learning-based Hybrid Recommendation Framework** for Anime and Movies. The system aims to extract meaningful patterns from user rating logs and content metadata to predict user satisfaction. The study aims to investigate the effectiveness of combining **Matrix Factorization (SVD)** for collaborative filtering with **TF-IDF/Cosine Similarity** for content-based filtering. Additionally, the research seeks to demonstrate that a hybrid machine learning approach is sufficient to achieve high prediction accuracy and user satisfaction while maintaining computational efficiency, avoiding the excessive complexity associated with deep neural networks..

B. *Motivation*

The motivation for this research arises from the rapid increase in digital content consumption. Platforms like Netflix, Crunchyroll, and IMDb generate millions of data points daily. Manual curation of recommendations is inefficient and unscalable. Although Deep Learning techniques (like Autoencoders) have been adopted for massive-scale industry recommenders, their application in smaller-scale or academic projects introduces challenges related to high computational cost and lack of interpretability. In contrast, classical **Hybrid Machine Learning** techniques offer a balanced solution: they provide the discovery of "hidden gems" through collaborative filtering and the stability of content matching through metadata analysis. Given the distinct but overlapping nature of Anime and Movie fanbases, a unified hybrid model provides a practical and scalable solution, motivating the adoption of this ML-based approach.

C. *Summary*

This research focuses on applying **Hybrid Machine Learning techniques** to analyze viewer preferences. By leveraging structured features (ratings) and unstructured text (plot summaries/genres), the proposed system performs rating prediction, similarity matching, and top-N item recommendation. The study emphasizes the use of **interpretable models**, demonstrating their capability to handle the "Sparsity" and "Cold Start" problems effectively. The findings of this research highlight that combining algorithms leads to better Root Mean Square Error (RMSE) scores and higher user relevance compared to single-algorithm baselines.

TABLE I. COMPARATIVE REVIEW OF RECOMMENDATION SYSTEMS

Sr. No.	Paper Title	Citation (APA Format)	Algorithms Used	Reason for Choosing the Algorithm	Limitations	Key Findings
1	Hybrid Recommender Systems: Survey and Experiments	Burke, R. User modeling and user-adapted interaction.	Hybrid Taxonomy (Weighted, Switching, Mixed)	Foundational study on combining recommenders	predates modern ML libraries	Hybrids mitigate cold start and sparsity better than single methods.
2	Hybrid Recommendation System for Movies Using ANN	Sharma, et al.	SVD-CF, Neural Networks	Optimizes final ranking using AI techniques	High computational requirement	Achieved high accuracy and F1-score; reduced RMSE compared to baselines.
3	Hybrid Approach to Build Anime Recommendation System	Shardul Rudraksha, Swapnil Misa	Cascade Hybrid (CF + CBF)	Refines results sequentially for better precision	Requires dense user data for the CF stage	Hybrid method was found to be quicker and more accurate for anime.
4	Efficient Hybrid Movie Recommendation Framework	Sanjay Jain, Manash Sarkar	TF-IDF, Weighted Average, Min-Max Scaling	Captures both popularity and semantic similarity	Static model; lacks real-time adaptation	Improved relevance by combining popularity signals with content tags.
5	Hybrid Movie Recommendation System	Sonika Sharma IJERT.	TF-IDF, RoBERTa, Collaborative Filtering	Uses embeddings for deeper content understanding	RoBERTa is computationally heavy	Successfully addressed lack of diversity in recommendations.
6	Survey on Collaborative Filtering, Content-based Filtering and Hybrid Recommendation System	Poonam B. Throat, R. M. Goudar	Neighborhood-based CF, Metadata Similarity	Simple, stable implementation	Scalability issues with very large user matrices	Integrating CF and CBF provided more stable results on sparse data.
7	Deep Anime Recommendation System	Varun Prakash, Mrinal Pandey	Deep Learning + Hybrid Filtering	Learns complex non-linear user representations	High complexity for implementation	Outperformed baseline models in user satisfaction metrics.
8	Anime Recommendation System Using Machine Learning	Bhaskaruni Harini, T. kamaleshwar	Hybrid ML (CF + CBF)	Adapts to evolving user interests	Dependent on consistent user interaction history	More responsive to changing preferences than static lists.

II. Implementation

This section describes the practical implementation of the proposed **Hybrid Anime and Movie Recommendation System**. The implementation follows a structured pipeline consisting of data collection, preprocessing, feature engineering (content and collaborative), model building, and evaluation. In accordance with the project objectives, the system employs **classical machine learning techniques** (SVD, Cosine Similarity) to ensure efficiency and explainability.

A. Dataset Description

The dataset used in this implementation is a combination of standard benchmark datasets (such as **MovieLens**) and Anime datasets (such as **MyAnimeList** from Kaggle).

Movies Data: Includes attributes such as Movie ID, Title, Genres, Overview/Synopsis, and Cast.

Anime Data: Includes Anime ID, Title, Genres, Type (TV/Movie), Episodes, and Synopsis.

Ratings Data: A matrix of User IDs, Item IDs, and Ratings (scaled 1-10 or 1-5). The raw data is often messy, containing missing synopsis values or duplicate entries, making preprocessing a critical step.

B. Data Preprocessing

Data preprocessing is performed to ensure the quality of the input.

Text Cleaning: Removal of stopwords and special characters from the "Overview" and "Genre" fields to prepare for NLP tasks.

Merging: Integrating Movie and Anime databases into a unified structure where possible (e.g., mapping genres to a common list).

Filtering: Users with very few ratings are filtered out to reduce noise and improve the Collaborative Filtering model's performance.

C. Feature Selection and Engineering

Feature extraction focuses on deriving meaningful attributes for both filtering methods.

Content Features: We use **TF-IDF (Term Frequency-Inverse Document Frequency)** Vectorization on the "Synopsis" and "Genre" tags. This converts text data into numerical vectors, allowing us to calculate the similarity between two items.

Collaborative Features: We construct a **User-Item Matrix**, which represents the interaction history. Matrix Factorization techniques decompose this matrix to find latent features (hidden patterns) connecting users and items.

D. Machine Learning Model Implementation

he system implements a **Weighted Hybrid Strategy**:

Content-Based Component: Uses **Cosine Similarity** on the TF-IDF vectors to find items similar to those the user has highly rated.

Collaborative Component: Uses **SVD (Singular Value Decomposition)** or **K-Nearest Neighbors (KNN)** to predict ratings based on similar users.

Hybrid Engine: The final recommendation score is a weighted average of the Content Score and the Collaborative Score.

Formula:

$FinalScore = (\alpha \times CF_{Score}) + ((1 - \alpha) \times CBF_{Score})$ This approach is selected over deep learning due to its mathematical transparency and ease of tuning.

E. Model Evaluation

The analytical results are evaluated using metrics such as **RMSE (Root Mean Square Error)** for rating prediction accuracy. Visual outputs include:

Top-N Recommendation Lists: Personalized lists for specific users.

Correlation Heatmaps: Showing similarity between different genres or items.

Word Clouds: Visualizing the most frequent terms in the top recommended items.

F. Explainability and Interpretation

The final output provides statistically supported insights. By integrating content metadata, the system can explain *why* an item was recommended (e.g., "Recommended because you watched 'Naruto' and this item shares the genre 'Shonen'"). This maintains the interpretability that is often lost in "black box" deep learning models.

.